Jaeyeon Kim

Curriculum Vitae

Education

2024–Present **Ph.D. student in Computer Science**, *Harvard University, MA, USA.*
Advisor: Sitan Chen, Sham Kakade

2020–2024 **B.S. in Mathematics**, *Seoul National University, Seoul, South Korea.*
GPA: 4.23/4.30, Major GPA: 4.30/4.30

Research Interest

Diffusion Models, Discrete Diffusion

Publications

(* denotes co-first authors)

Conference

[1] *Stop Training for the Worst: Progressive Unmasking Accelerates Masked Diffusion Training*, **J. Kim***, J. Geuter, D. Alvarez-Melis, S. Kakade, S. Chen, ***ICML 2026***.

[2] *Discrete Tilt Matching*, Y. Chen*, S. Wang*, P. Potaptchik, **J. Kim**, M. Albergo, ***ICML 2026***.

[3] *Fine-Tuning Masked Diffusion for Provable Self-Correction*, **J. Kim***, S. Kim*, T. Lee*, D. Z. Pan, H. Kim, S. Kakade, S. Chen, ***ICML 2026.***

[4] *Delta Velocity Rectified Flow for Text-to-Image Editing*, G. Beaudouin, M. Li, **J. Kim**, S. Yoon, M. Wang, ***CVPR 2026***.

[5] *Any-Order Flexible Length Masked Diffusion*, **J. Kim***, L. Cheuk-Kit*, C. Domingo-Enrich, Y. Du, S. Kakade, T. Ngotiaoco, S. Chen, M. Albergo, ***ICLR 2026.***

[6] *Train for the Worst, Plan for the Best: Understanding Token Ordering in Masked Diffusions*, **J. Kim***, K. Shah*, V. Kontonis, S. Kakade, S. Chen, ***ICML 2025 (Oral),* Outstanding paper.**

[7] *LoRA Training Provably Converges to a Low-Rank Global Minimum or It Fails Loudly (But it Probably Won't Fail)*, J. Kim, **J. Kim**, E. K. Ryu, ***ICML 2025 (Oral)***.

[8] *Task Diversity Shortens the ICL Plateau*, **J. Kim***, S. Kwon*, J. Choi, J. Park, J. Cho, J. D. Lee, E. K. Ryu, ***TMLR***.

[9] *Optimal Acceleration for Minimax and Fixed-Point Problems is Not Unique*, T. Yoon, **J. Kim**, J. J. Suh, E. K. Ryu, ***ICML 2024 (Spotlight)***.

[10] *Time-Reversed Dissipation Induces Duality Between Minimizing Gradient Norm and Function Value*, **J. Kim**, A. Ozdaglar, C. Park, E. K. Ryu, ***NeurIPS 2023***.

✉ *jaeyeon_kim@g.harvard.edu*

Preprint

[11] *Selective Underfitting in Diffusion Models*, K. Song*, **J. Kim***, S. Chen, Y. Du, S. Kakade, V. Sitzmann ***Under review at NeurIPS 2026***

[12] *Mirror Duality in Convex Optimization*, **J. Kim**, C. Park, A. Ozdaglar, J. Diakonikolas, E. K. Ryu, **Under review at SiOPT**.

[13] *What Does Fréchet Distance Measure? A Directional Decomposition*, Y. Lee, **J. Kim**, ***arXiv preprint, 2026***.

[14] *Blackboard Intelligence Can Surpass Autoregressive on Globally Constrained Problems*, W. Jeon*, **J. Kim***, S. Kakade, Y. Du, A. S. Bedi, A. K. Chithanar, C. Lee, T. Kim, S. Chen, ***arXiv preprint, 2026***.

[15] *Retrofitting Linear Attention into Diffusion Language Models*, J. Kim, Y. Roh, **J. Kim**, ***arXiv preprint, 2026***.

[16] *From Interface to Inference: Eliciting Any-Order Inference from Any-Order Models*, S. Kim*, **J. Kim***, T. Lee*, Y. Chen*, Y. Du, S. Kakade, S. Chen, ***arXiv preprint, 2026***.

[17] *Rethinking Self-Consistency in Protein Generative Models*, M. Lee*, **J. Kim***, Y. Lin, M. AlQuraishi, ***ICML 2026 Workshop on Generative and Agentic AI for Biology***.

Awards

2026- **Kempner Graduate Fellow**.
- Three-year fellowship supporting PhD study.

2026- **Jump Trading Fellowship**.
- One of eight PhD students selected nationwide; fellowship includes full tuition and a $50,000 stipend. [Link]

2025 **Outstanding Paper Award**, ICML 2025.
- One of six papers selected for the Outstanding Paper Award from approximately 15,000 submissions. [Link]

2024–Present **Ilju Scholarship**.
- One of six PhD students selected nationwide in South Korea; provides $24,000 in annual funding for up to five years of study.

2021 **3rd Place**, Simon Marais Mathematics Competition.
- Mathematics competition for university students in Asia

2020, 2021 **Gold Medal**, Korean Undergraduate Mathematics Competition.
- Mathematics competition for university students in Korea

2020–2023 **Presidential Science Scholarship**, Korea Student Aid Foundation.
- 8k in annual financial support

2019 **Silver Medal (11th Place)**, Romanian Masters of Mathematics.
- International mathematics competition for pre-university students

2018–2019 **Korean Finalist for the International Mathematical Olympiad (IMO)**.
- Final-round students (13 students)

Talk / Services

Nov. 2026 **Invited talk (scheduled)**, *SIAM Conference on Mathematics of Data Science*, Salt Lake City, Utah.

Aug. 2026 **Invited talk**, *Generative Modeling Seminar*, Microsoft Research New England.

2026 **Main organizer**, *NonAR-LM Workshop at COLM 2026*.
- Served as a main organizer of the first workshop on diffusion language models. [Link]

Reviewer, *ICML 2025, ICLR 2026, ICML 2026, NeurIPS 2026*.

Experience

2026 Summer **Engineering Intern**, Autopilot team, Tesla AI.
- Worked on text reasoning for self-driving models.

2022-2024 **Research Intern**, under Ernest Ryu, *Seoul National University*.
- Worked on optimization theory

2023 Summer **Research Intern**, under Asuman Ozdaglar, *MIT (remote)*.
- Worked on optimization theory

Teaching experiences

2026 Spring Foundation of Reinforcement Learning (CS2824), *Harvard University*

2025 Fall Computational Optimal Transport (CS2840), *Harvard University*

2024 Spring Machine Learning Theory, *Seoul National University*

2022 Fall Mathematical Foundation of Deep Neural Networks, *Seoul National University*

Activities

2025–Present Vice President of the Harvard Graduate Korean Society, *Harvard University*

2022–2023 Vice President of the Global Leadership Program, *Seoul National University*

2021–2022 Vice President of the Student Association in the Department of Mathematics, *Seoul National University*

Languages

English **Fluent**

Korean **Native**